

Problem 1. Averaging argument.

In the lecture, we have seen the following version of the “No free lunch” theorem:

Theorem 1. *Consider binary classification with 0–1 loss and $\mathcal{X}$ infinite. For any $m > 0$ and any learning algorithm A , there exists a probability distribution $\mathcal{D}$ such that*

$$\mathcal{R}_\star = 0, \quad \mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}(A(S))] \geq \frac{1}{4}.$$

You will prove the following corollary using [Theorem 1](#):

Corollary 2. *Let $\mathcal{X}$ be infinite and let $\mathcal{H}$ be the set of all functions from $\mathcal{X}$ to $\{0, 1\}$. Then $\mathcal{H}$ is not PAC-learnable.*

Solution.

Suppose, toward a contradiction, that $\mathcal{H}$ is PAC-learnable. Then there are a learning algorithm A and a finite function $m_{\mathcal{H}}(\epsilon, \delta)$ such that for every distribution $\mathcal{D}$ realizable by $\mathcal{H}$ and every integer $m \geq m_{\mathcal{H}}(\epsilon, \delta)$,

$$\Pr_{S \sim \mathcal{D}^m} \{ \mathcal{R}_{\mathcal{D}}(A(S)) > \epsilon \} \leq \delta,$$

where $\mathcal{R}_{\mathcal{D}}(h) = \Pr_{(X,Y) \sim \mathcal{D}} \{ h(X) \neq Y \}$.

Fix $\epsilon = \delta = 1/8$, and choose an integer $m \geq \max\{1, m_{\mathcal{H}}(1/8, 1/8)\}$. By [Theorem 1](#), there exists a distribution $\mathcal{D}$ with zero Bayes risk and $\mathbb{E}_{S \sim \mathcal{D}^m}[\mathcal{R}_{\mathcal{D}}(A(S))] \geq 1/4$. Because $\mathcal{H}$ contains every binary classifier, the zero-risk target for this distribution belongs to $\mathcal{H}$; hence $\mathcal{D}$ is realizable by $\mathcal{H}$.

Set $Z = \mathcal{R}_{\mathcal{D}}(A(S))$ and $p = \Pr\{Z > 1/8\}$. The 0–1 loss implies $0 \leq Z \leq 1$. Splitting the expectation according to whether $Z \leq 1/8$ gives the averaging bound

$$\begin{aligned} \frac{1}{4} &\leq \mathbb{E}[Z] = \mathbb{E}[Z \mathbf{1}_{\{Z \leq 1/8\}}] + \mathbb{E}[Z \mathbf{1}_{\{Z > 1/8\}}] \\ &\leq \frac{1}{8} \Pr\{Z \leq 1/8\} + \Pr\{Z > 1/8\} = \frac{1}{8} + \frac{7}{8}p. \end{aligned}$$

Rearranging yields

$$\Pr_{S \sim \mathcal{D}^m} \left\{ \mathcal{R}_{\mathcal{D}}(A(S)) > \frac{1}{8} \right\} = p \geq \frac{1/4 - 1/8}{7/8} = \frac{1}{7} > \frac{1}{8}.$$

But the PAC guarantee with $\epsilon = \delta = 1/8$ requires this probability to be at most $1/8$, a contradiction. The distribution may depend on A and m , since the PAC guarantee must hold uniformly over all realizable distributions. Therefore $\mathcal{H}$ is not PAC-learnable. $\square$

Problem 2. ERM for learning closed intervals.

Let $\mathcal{X} = \mathbb{R}$ and $\mathcal{H}$ be the set of indicator functions of closed intervals and the empty set:

$$\mathcal{H} = \{\mathbf{1}_{a \leq x \leq b} : \forall a, b \in \mathbb{R}\}.$$

Assume binary classification with 0–1 loss and realizability. Prove that $\mathcal{H}$ can be PAC-learned using ERM with sample complexity

$$O\left(\frac{\log(1/\delta)}{\epsilon}\right).$$

You need to provide a description of the learning algorithm and prove its sample complexity and the PAC learning guarantee.

Solution.

Learning algorithm and the ERM property. Let $S = ((x_1, y_1), \dots, (x_m, y_m))$, and set $J = \{i : y_i = 1\}$. If $J = \emptyset$, output the empty-set classifier $\hat{h} \equiv 0$. Otherwise, define

$$L = \min_{i \in J} x_i, \quad U = \max_{i \in J} x_i, \quad \hat{h}(x) = \mathbf{1}_{[L, U]}(x).$$

Thus the learner takes the smallest closed interval containing all positive examples.

Fix an arbitrary realizable distribution $\mathcal{D}$, and let μ be its marginal distribution on $\mathbb{R}$. There is a target $h_\star \in \mathcal{H}$ with $\mathcal{R}_{\mathcal{D}}(h_\star) = 0$. With probability one, all labels in the sample agree with $h_\star$; we work on this probability-one event throughout the proof. If the target is the empty set, every sample label is zero and the learner has risk zero. Otherwise, write $h_\star = \mathbf{1}_{[a, b]}$ with $a \leq b$. When $J \neq \emptyset$, every positive example lies in $[a, b]$, so

$$a \leq L \leq U \leq b.$$

All positive examples lie in $[L, U]$, and no negative example can lie in this interval, since $[L, U] \subseteq [a, b]$. When $J = \emptyset$, the empty-set classifier also labels every sample point correctly. In either case,

$$\widehat{\mathcal{R}}_S(\hat{h}) := \frac{1}{m} \sum_{i=1}^m \mathbf{1}_{\{\hat{h}(x_i) \neq y_i\}} = 0 = \min_{h \in \mathcal{H}} \widehat{\mathcal{R}}_S(h).$$

Hence this learner is an ERM on every realizable sample.

On arbitrary samples, if the proposed interval contains a negative example, instead minimize empirical risk over the empty set and all $[x_i, x_j]$ with $x_i \leq x_j$. This finite set represents every interval's predictions on the sample: an interval including sample points can be replaced by the interval from their smallest to largest coordinate without changing those predictions. Thus this fallback is an ERM over $\mathcal{H}$; it is never needed under realizability.

Risk of the output. Since the learned interval is contained in the target, the learner makes only false-negative errors. Put $p = \mu([a, b])$. If there are no positive examples, its risk is p ; otherwise,

$$\mathcal{R}_{\mathcal{D}}(\hat{h}) = \mu([a, L]) + \mu((U, b]). \tag{1}$$

In particular, the risk is always at most p . If $p \leq \epsilon$, the desired error guarantee therefore holds for every sample in the probability-one event above. It remains to consider $p > \epsilon$.

Boundary regions, including distributions with atoms. Set $t = \epsilon/2$. We claim there exist $c, d \in [a, b]$ satisfying

$$\begin{aligned}\mu([a, c)) &\leq t \leq \mu([a, c]), \\ \mu((d, b]) &\leq t \leq \mu([d, b]).\end{aligned}\tag{2}$$

To justify this carefully, define

$$c = \inf\{x \in [a, b] : \mu([a, x]) \geq t\}.$$

The set is nonempty because $\mu([a, b]) = p > t$, and its infimum lies in $[a, b]$. For every $x < c$ in $[a, b]$, we have $\mu([a, x]) < t$. If $c > a$, take such x increasing to c ; continuity of a measure from below gives $\mu([a, c)) \leq t$. If $c = a$, this inequality holds because $[a, c)$ is empty. If $c < b$, every $x \in (c, b]$ satisfies $\mu([a, x]) \geq t$: by the definition of the infimum, there is a point of the defining set strictly below x , and monotonicity then applies. Taking x decreasing to c and using continuity from above gives $\mu([a, c]) \geq t$. If $c = b$, the same inequality follows from $\mu([a, b]) = p > t$. Applying this argument with the order reversed, equivalently to the reflected variable $-X$, gives

$$d = \sup\{x \in [a, b] : \mu([x, b]) \geq t\}$$

and the second pair of inequalities in (2). This construction does not assume that μ has a density or that one can find regions with mass exactly t .

Let $B_L = [a, c]$ and $B_R = [d, b]$. Both regions have probability at least $t = \epsilon/2$. Suppose the sample contains a point in each region. Both of these observations are positive, so $J \neq \emptyset$, $L \leq c$, and $U \geq d$. Therefore (1) and (2) imply

$$\mathcal{R}_{\mathcal{D}}(\hat{h}) \leq \mu([a, c)) + \mu((d, b]) \leq 2t = \epsilon.$$

The regions need not be disjoint; one observation may hit both. In particular, this event also handles the possibility that the sample would otherwise contain no positive examples.

Failure probability and sample complexity. For any measurable set B , independence gives $\Pr\{x_i \notin B \text{ for all } i\} = (1 - \mu(B))^m$. Failure of the error guarantee can occur only if at least one of the two boundary regions is missed. A union bound and $1 - u \leq e^{-u}$ for $u \in [0, 1]$ now give

$$\begin{aligned}\Pr_{S \sim \mathcal{D}^m} \{\mathcal{R}_{\mathcal{D}}(\hat{h}) > \epsilon\} &\leq (1 - \mu(B_L))^m + (1 - \mu(B_R))^m \\ &\leq 2(1 - \epsilon/2)^m \leq 2e^{-m\epsilon/2}.\end{aligned}$$

Thus, for every $0 < \epsilon, \delta < 1$, it suffices to take

$$m \geq \left\lceil \frac{2}{\epsilon} \log \frac{2}{\delta} \right\rceil$$

to ensure that the risk is at most ϵ with probability at least $1 - \delta$. Here $\log$ denotes the natural logarithm. The bound is uniform over all realizable distributions and targets, proving PAC learnability. In the customary confidence range $0 < \delta \leq 1/2$, the inequality $\log(2/\delta) \leq 2\log(1/\delta)$ gives the requested complexity $O(\log(1/\delta)/\epsilon)$. The explicit bound above remains valid for the entire range $0 < \delta < 1$. $\square$

Problem 3. VC dimensions.

Given some finite domain set, $\mathcal{X}$, and a number $k \leq |\mathcal{X}|$, figure out the VC dimension of each of the following classes (and prove your claims):

1. $\mathcal{H}_{=k}^{\mathcal{X}} = \{h \in \{0, 1\}^{\mathcal{X}} : |\{x : h(x) = 1\}| = k\}$. That is, the set of all functions that assign the value 1 to exactly k elements of $\mathcal{X}$.
2. $\mathcal{H}_{\text{at-most-}k} = \{h \in \{0, 1\}^{\mathcal{X}} : |\{x : h(x) = 1\}| \leq k \text{ or } |\{x : h(x) = 0\}| \leq k\}$.

Solution.

Write $n = |\mathcal{X}|$, and assume k is an integer with $0 \leq k \leq n$. Recall that $S \subseteq \mathcal{X}$ is *shattered* by a class if for every map $\ell : S \rightarrow \{0, 1\}$ there exists a hypothesis in the class whose restriction to S equals ℓ . The VC dimension is the largest cardinality of a shattered subset.

1. **Exactly k positive points.** We claim that

$$\text{VCdim}(\mathcal{H}_{=k}^{\mathcal{X}}) = \min\{k, n - k\}.$$

Upper bound. Suppose $S \subseteq \mathcal{X}$ is shattered and $|S| = d$. The labeling assigning 1 to every point of S must extend to a hypothesis with exactly k positive points. This forces $d \leq k$. The labeling assigning 0 to every point of S must also extend to such a hypothesis. Its k positive points must all lie in $\mathcal{X} \setminus S$, so $k \leq n - d$, or $d \leq n - k$. Consequently,

$$\text{VCdim}(\mathcal{H}_{=k}^{\mathcal{X}}) \leq \min\{k, n - k\}.$$

Lower bound. Set $d = \min\{k, n - k\}$ and choose any $S \subseteq \mathcal{X}$ of size d . Let ℓ be an arbitrary labeling of S , and put $T = \{x \in S : \ell(x) = 1\}$ and $r = |T|$. We need to add exactly $k - r$ positive points outside S . This is possible because

$$0 \leq k - r \leq k \leq n - d = |\mathcal{X} \setminus S|,$$

where $r \leq d \leq k$ gives the first inequality and $d \leq n - k$ gives the last one. Choose $V \subseteq \mathcal{X} \setminus S$ with $|V| = k - r$, and define

$$h(x) = \mathbf{1}_{T \cup V}(x).$$

Then h has exactly $r + (k - r) = k$ positive points and agrees with ℓ on S . Since ℓ was arbitrary, S is shattered, proving the matching lower bound.

2. **At most k points of either label.** We claim that

$$\text{VCdim}(\mathcal{H}_{\text{at-most-}k}) = \min\{n, 2k + 1\}.$$

Lower bound. Let $d = \min\{n, 2k + 1\}$ and choose any $S \subseteq \mathcal{X}$ with $|S| = d$. Consider any labeling $\ell : S \rightarrow \{0, 1\}$, and let r be its number of ones. At least one of $r \leq k$ and $d - r \leq k$ must hold. Indeed, if both failed, integrality would give

$$d = r + (d - r) \geq (k + 1) + (k + 1) = 2k + 2,$$

contradicting $d \leq 2k + 1$. If $r \leq k$, extend ℓ by assigning 0 to every point of $\mathcal{X} \setminus S$. The resulting hypothesis has at most k ones in the entire domain and therefore belongs to the class. If $d - r \leq k$, extend ℓ by assigning 1 to every point of $\mathcal{X} \setminus S$ instead; this hypothesis has at most k zeros in the entire domain. In either case there is an admissible extension. Thus S is shattered and

$$\text{VCdim}(\mathcal{H}_{\text{at-most-}k}) \geq \min\{n, 2k + 1\}.$$

Upper bound. The VC dimension is at most n , because every subset of $\mathcal{X}$ has at most n points. Now let $S \subseteq \mathcal{X}$ have $d \geq 2k + 2$ points, whenever such a subset exists. Label exactly $k + 1$ of its points by 1, and label the remaining $d - k - 1 \geq k + 1$ points by 0. Every extension of this labeling to $\mathcal{X}$ contains at least $k + 1$ ones and at least $k + 1$ zeros. It therefore satisfies neither of the two conditions defining $\mathcal{H}_{\text{at-most-}k}$. This labeling cannot be realized, so no set of size at least $2k + 2$ is shattered. Combining the two upper bounds gives

$$\text{VCdim}(\mathcal{H}_{\text{at-most-}k}) \leq \min\{n, 2k + 1\},$$

as required.

These formulas include the boundary cases. For $k = 0$ or $k = n$, the first class is a singleton and has VC dimension zero. For $k = 0$ and $n \geq 1$, the second class consists of the two constant functions and has VC dimension one. If $n \leq 2k + 1$, every labeling of $\mathcal{X}$ has at most k points of at least one label, so the second class contains all binary functions and has VC dimension n . If $n = 0$, both dimensions are zero. $\square$